

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

INDEX

Sr. No.	Description	Page No.
1.	Methodology	1
	1 Satellite SAR System with Resolution and Revisit Time	
	2 Retrieving Multi-Temporal SAR Imagery Over the Specified Period	
	3 Differential SAR Interferometry (D-InSAR) for Displacement Measurement	2
2.	Visualization	3
	1 Defining specific bridge location & parameters for satellite SAR based inspection	
	2 Calibration, georeferencing, orthorectification of SAR images (VH/VV)	4
	3 Results of SAR based bridge inspection, including visualizations	5
	3.1 VV Displacement From 23 Jan 2025 to 28 June 2025	6
	3.2 VH Displacement From 23 Jan 2025 to 28 June 2025	6
	3.3 SAR Based Analysis at Abutment 1 LHS	7
	3.4 SAR Based Analysis at Abutment 1 RHS	8
	3.9 SAR Based Analysis at Abutment 2 LHS	9
	3.10 SAR Based Analysis at Abutment 2 RHS	10
3.	Analysis	11
	1 Identify potential deformation, shifts or movement of the bridge structure	
	Conclusion	

METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
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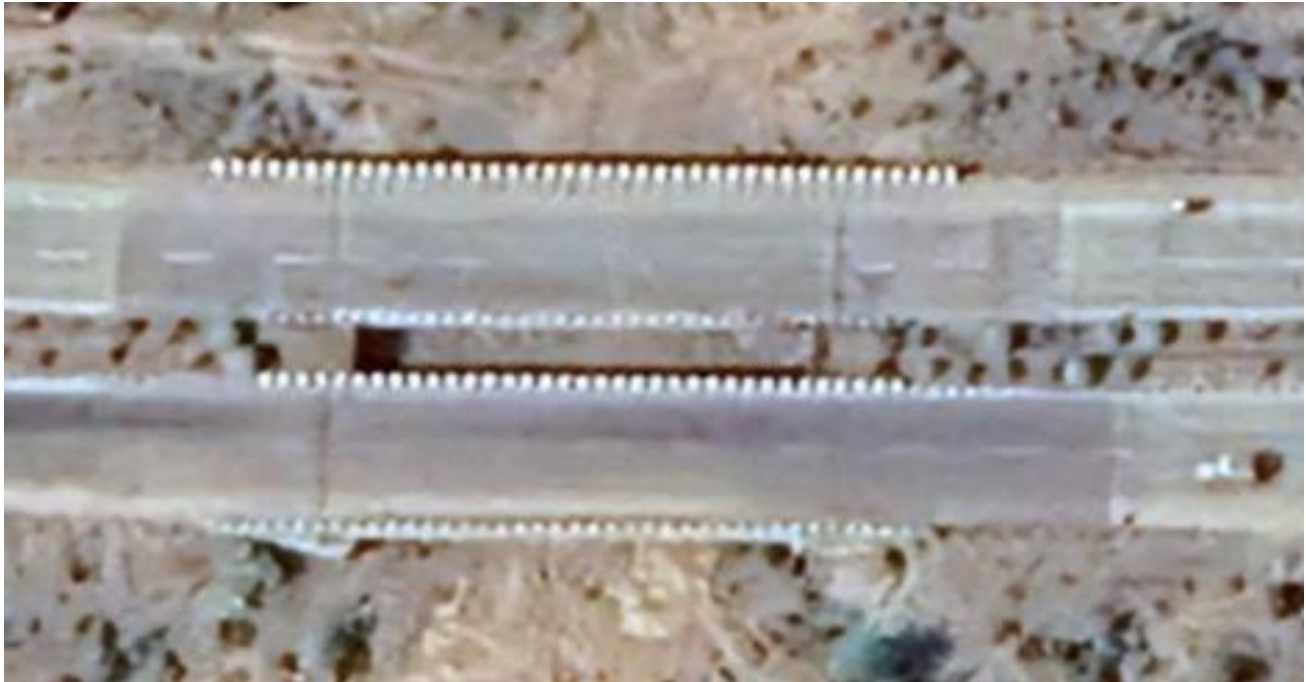
3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 969+900

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

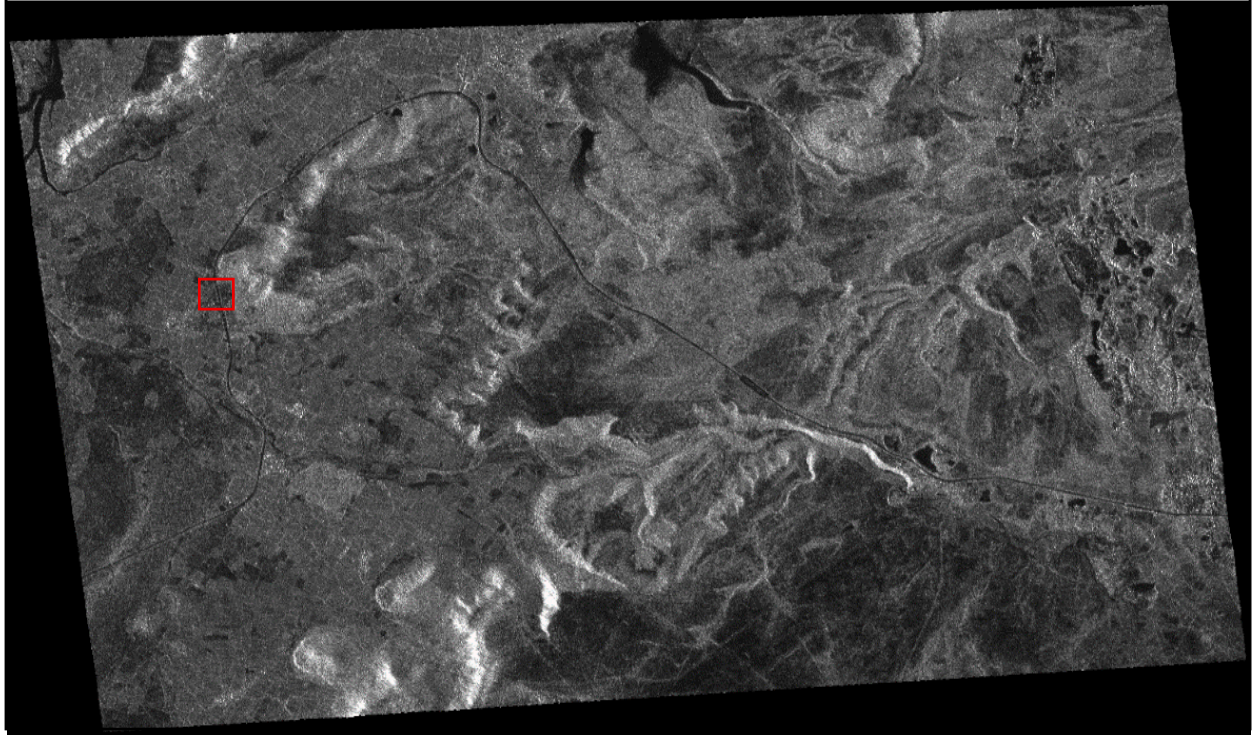
Flight Direction - DESCENDING

Polarization – VV+VH

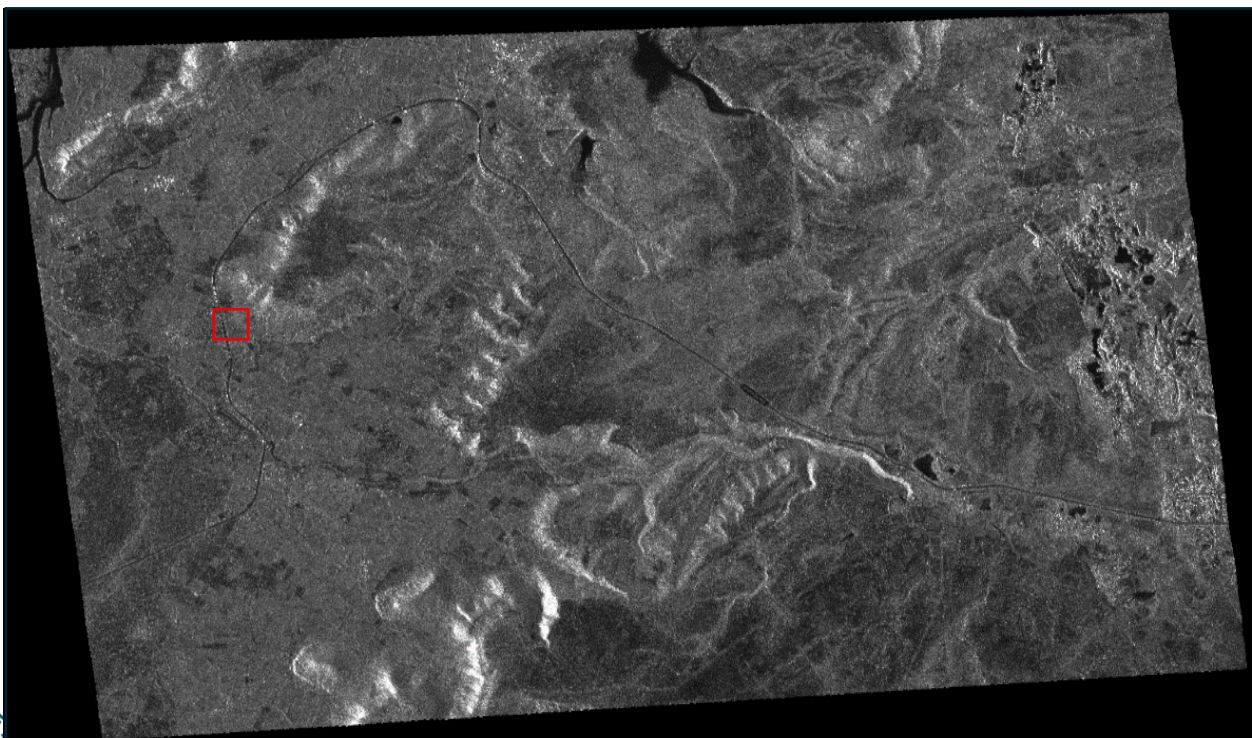
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

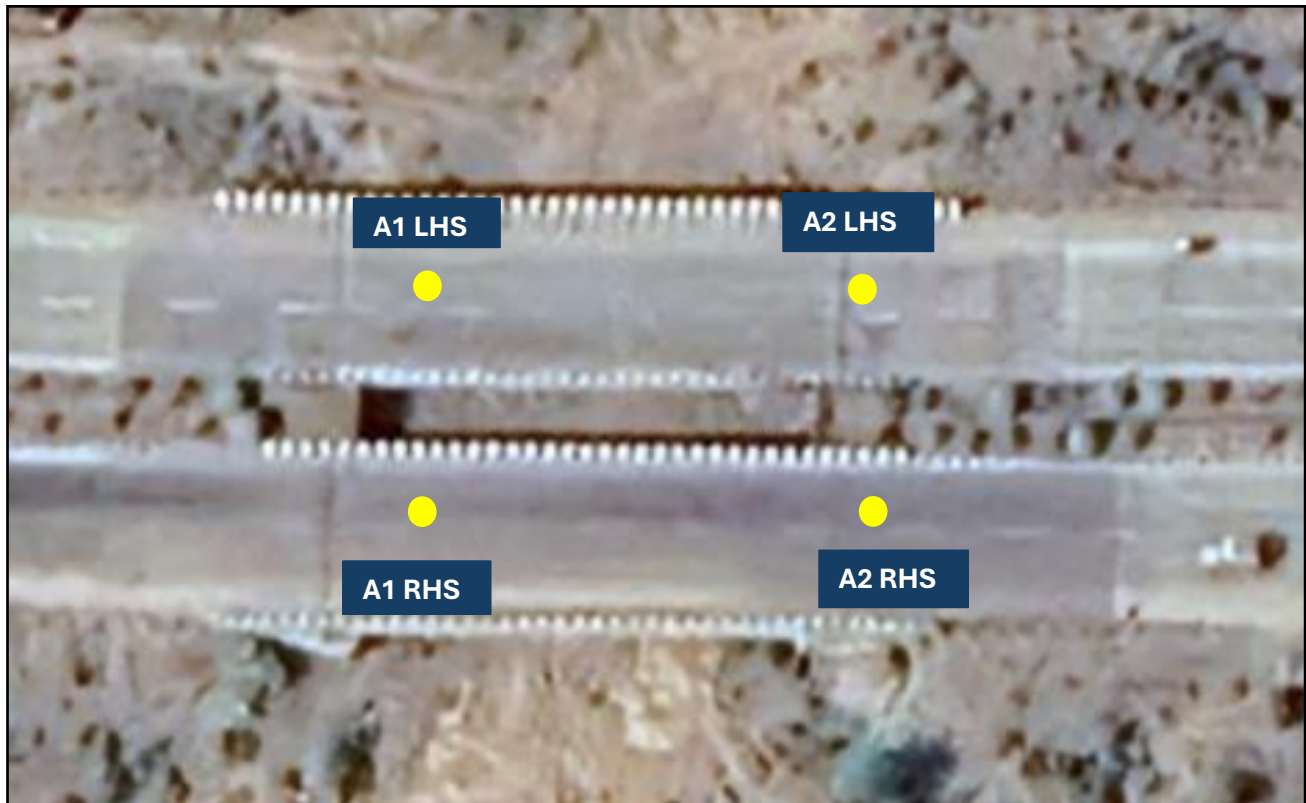


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

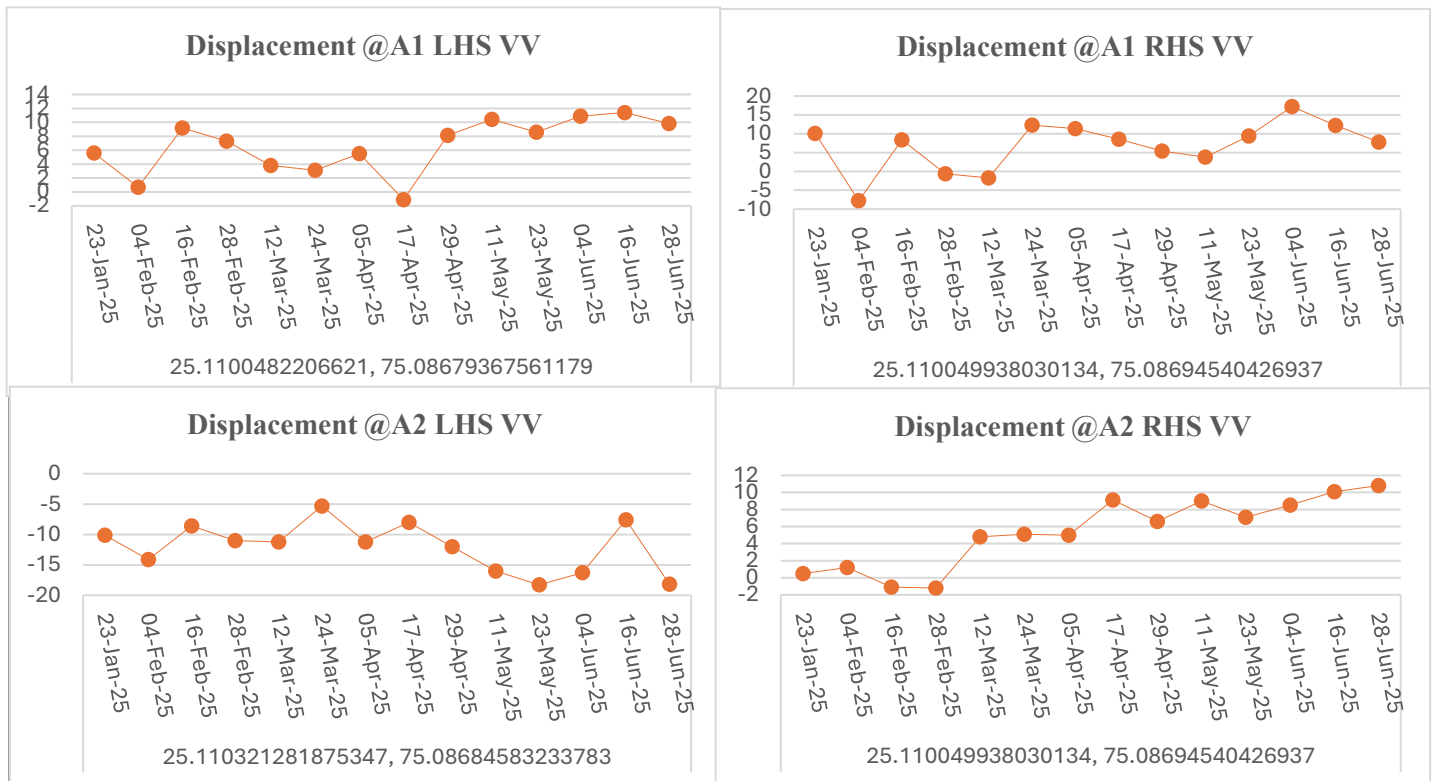
MNB – CH-969+900

Length – 43.21M

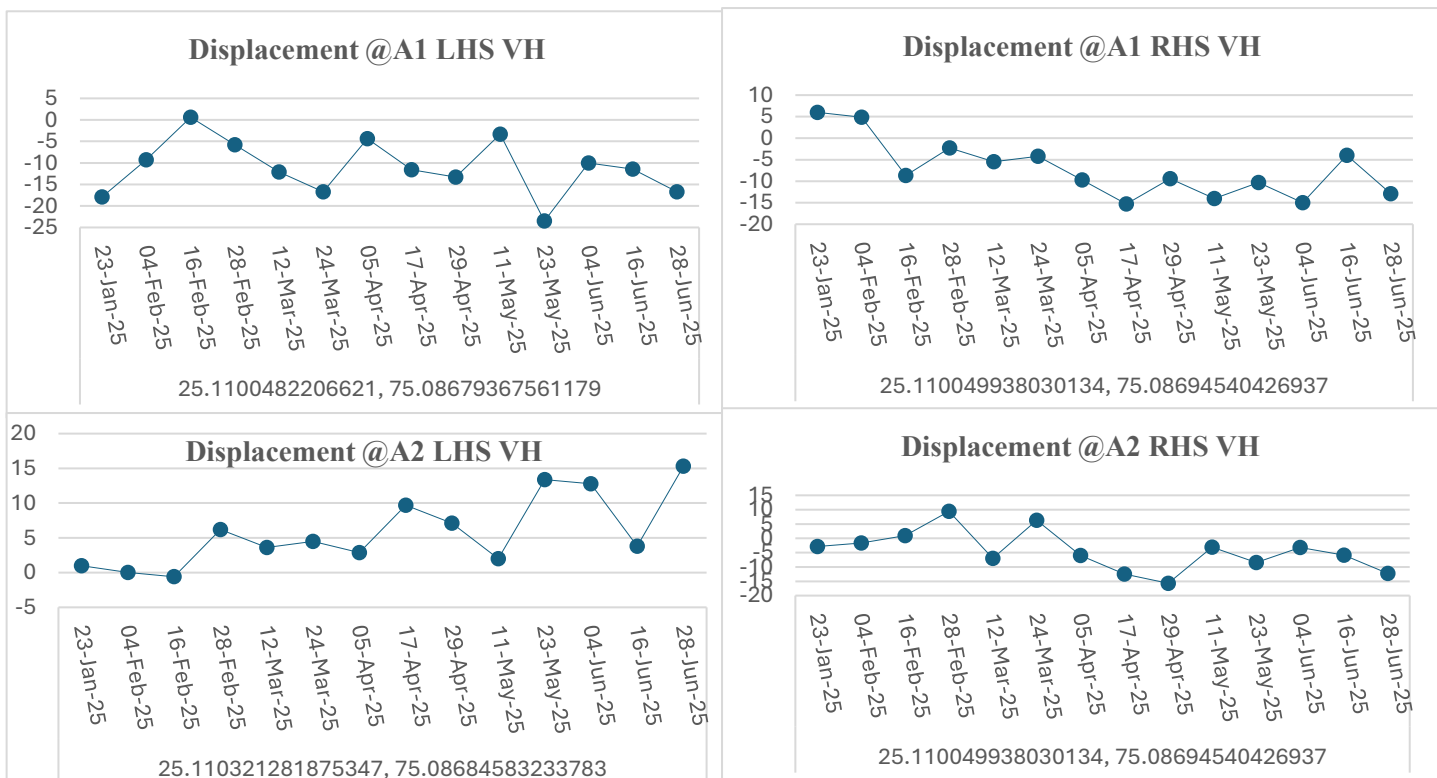


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

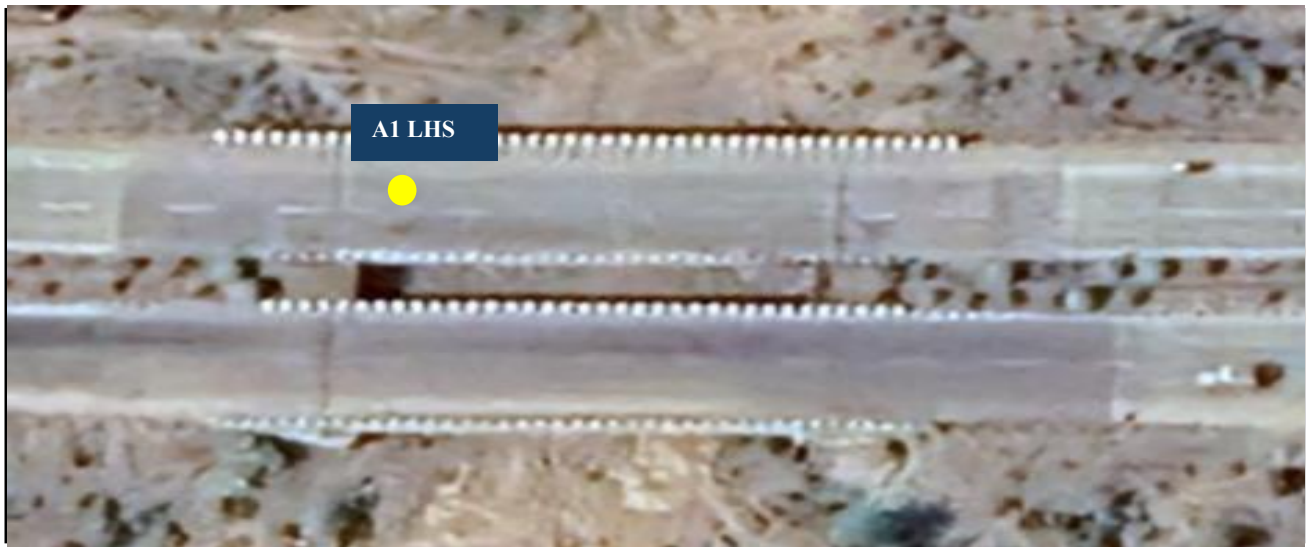
3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



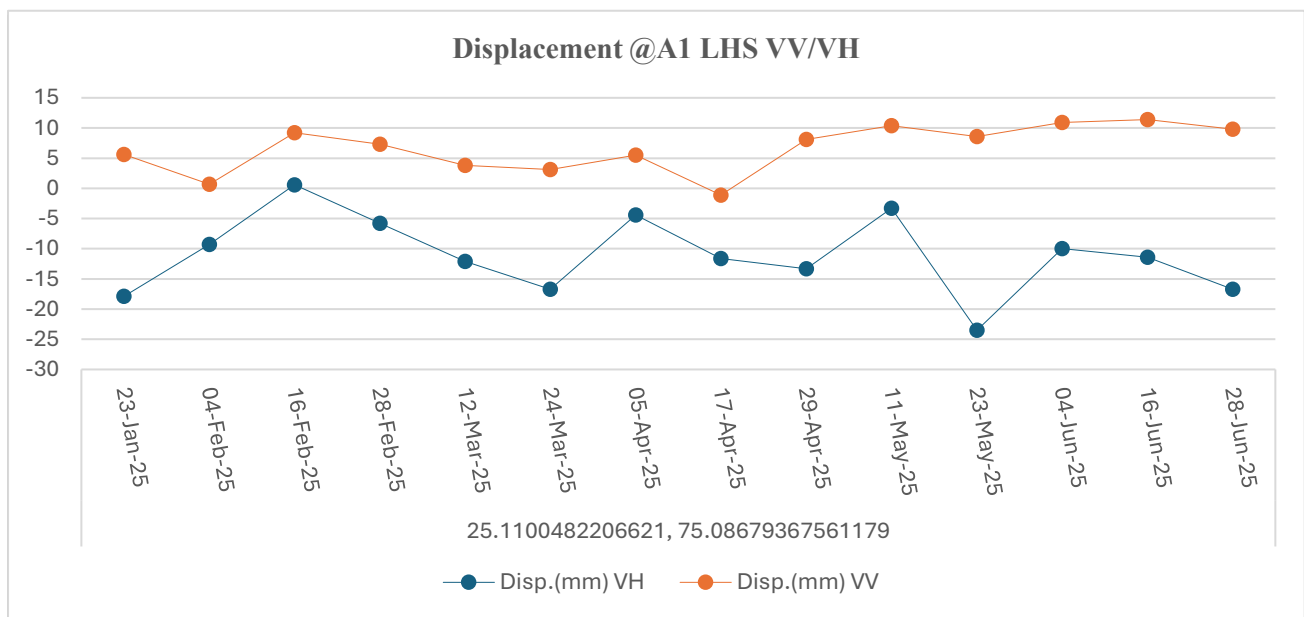
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.1100482206621, 75.08679367561179)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement remains consistently positive and moderately stable, generally fluctuating between +2 mm and +12 mm throughout the monitoring period.
- **Horizontal displacement (VH)** Horizontal displacement is predominantly negative, showing significant variability across the period.

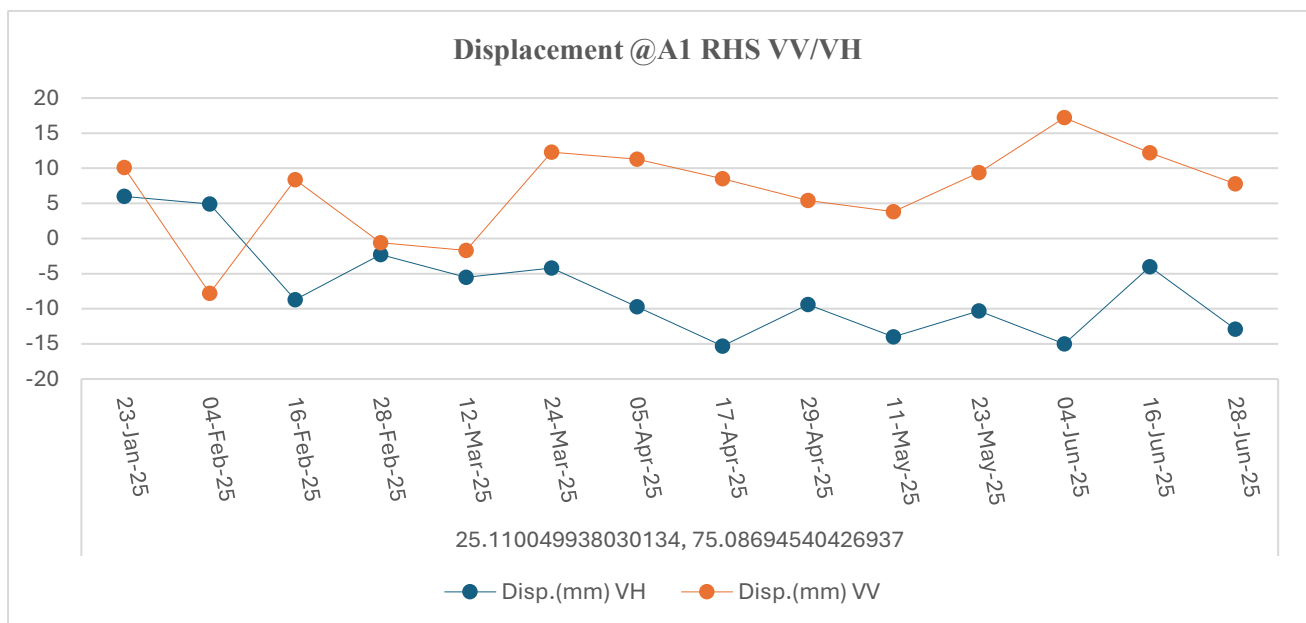
Critical Events & Anomalies:

- **February 2025:** A short-lived reduction in negative VH displacement (around -3 to -5 mm) is observed, suggesting temporary lateral relaxation or load redistribution.
- **April-May2025:** A sharp negative spike in VH occurs around 23 May 2025, reaching approximately -25 mm, representing the maximum lateral displacement recorded in the monitoring period.
- **March-June 2025:** VV shows relatively stable elevated values (+8 to +12 mm) while VH continues fluctuating at higher negative magnitudes, indicating combined vertical stability with lateral movement dominance.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.110049938030134, 75.08694540426937)



Key Observations:

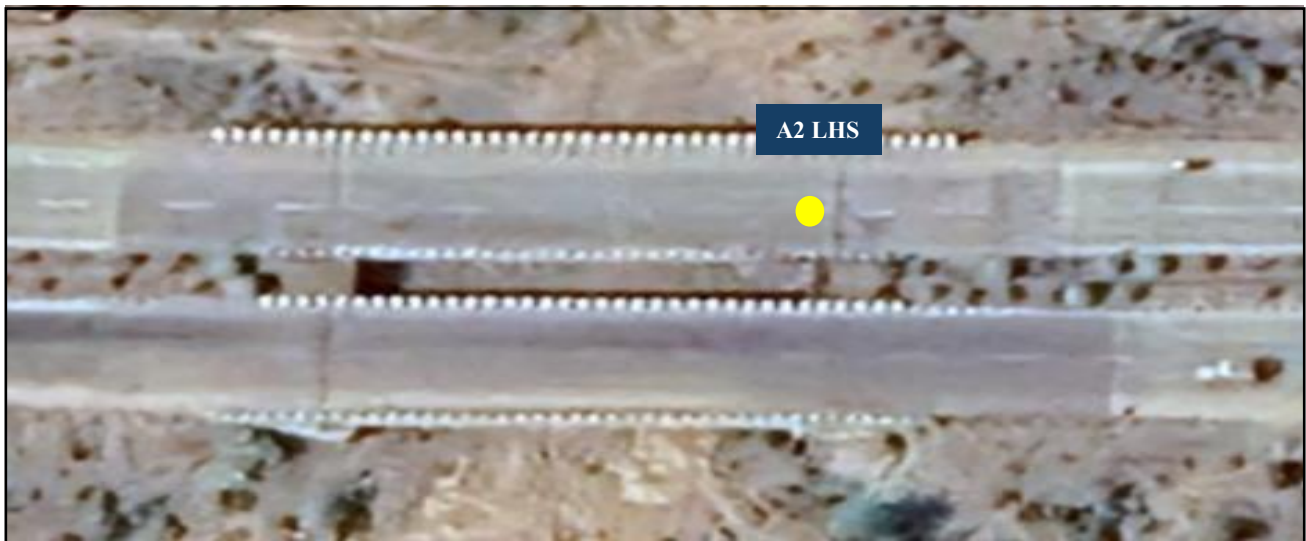
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement is predominantly negative, indicating sustained lateral movement in a consistent direction.
- **Vertical displacement (VV)** Vertical displacement exhibits moderate variability, generally ranging between -8 mm and $+18$ mm over the monitoring period (Jan–Jun 2025).

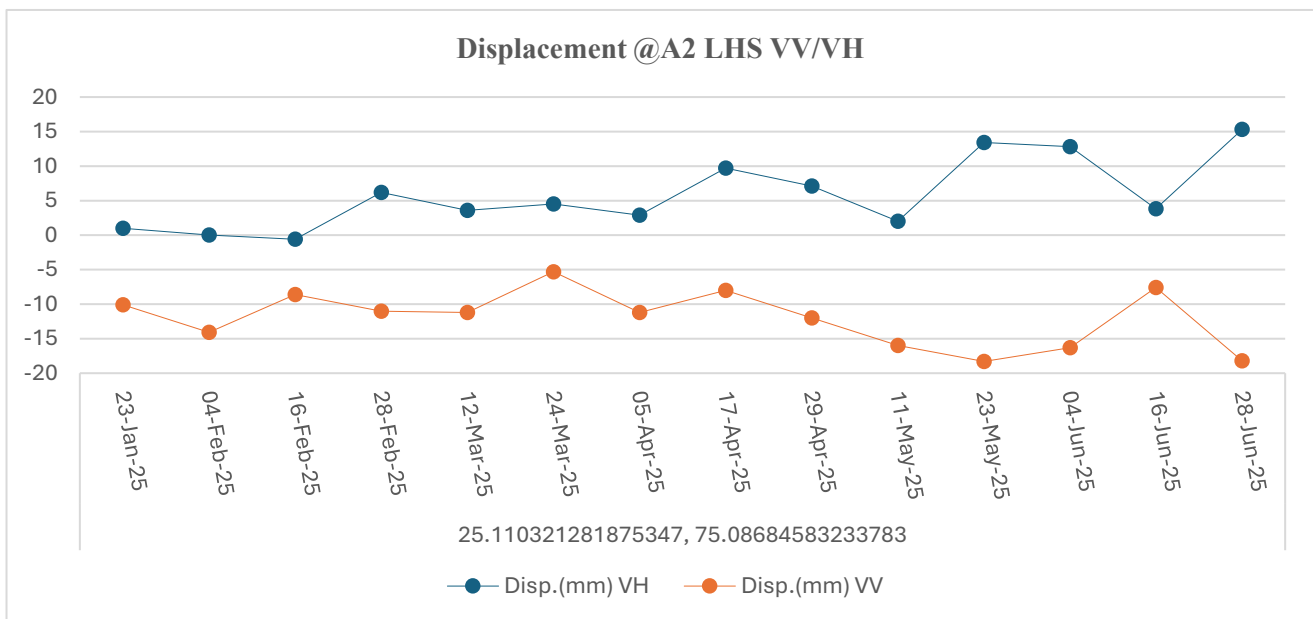
Critical Events & Anomalies:

- **February 2025:** A sharp negative VV excursion (around -8 mm) coincides with increasing negative VH displacement, indicating a short-term combined vertical and lateral response.
- **June 2025:** VV records its highest uplift peak ($+18$ mm), while VH remains strongly negative, highlighting decoupled vertical uplift with continued lateral displacement

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.110321281875347, 75.08684583233783)



Key Observations:

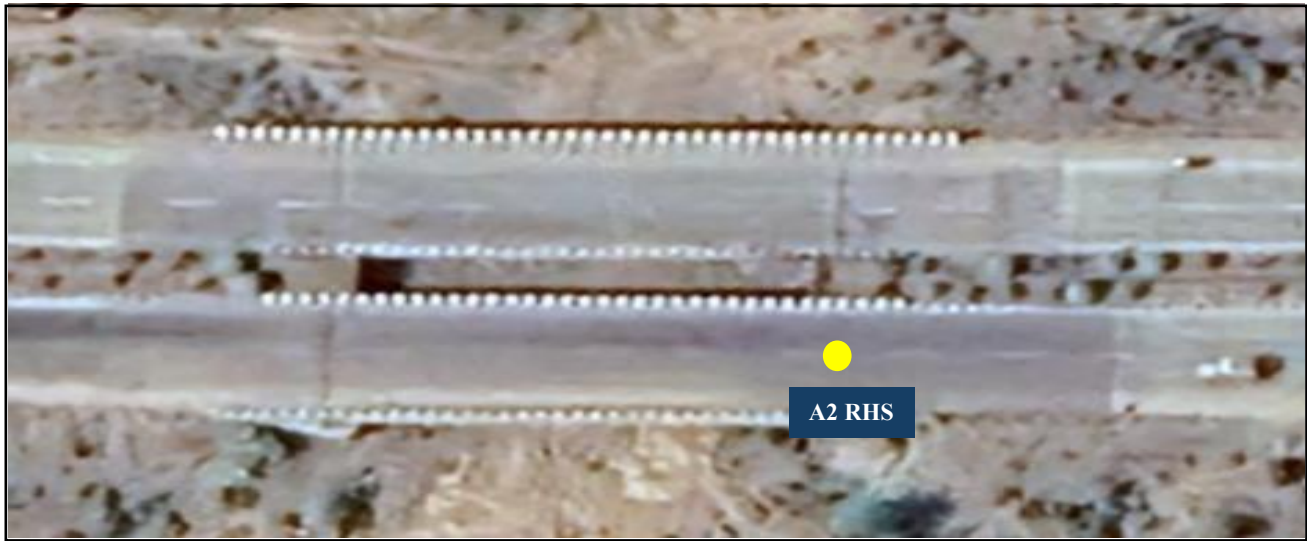
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement remains consistently negative, indicating dominant downward movement throughout the monitoring period.
- **Horizontal Displacement (VH)** Horizontal displacement is predominantly positive, showing moderate to high variability. VH values fluctuate approximately between 0 mm and +15 mm. An overall upward lateral trend is evident from February onwards, with several short-term oscillations but no prolonged reversals

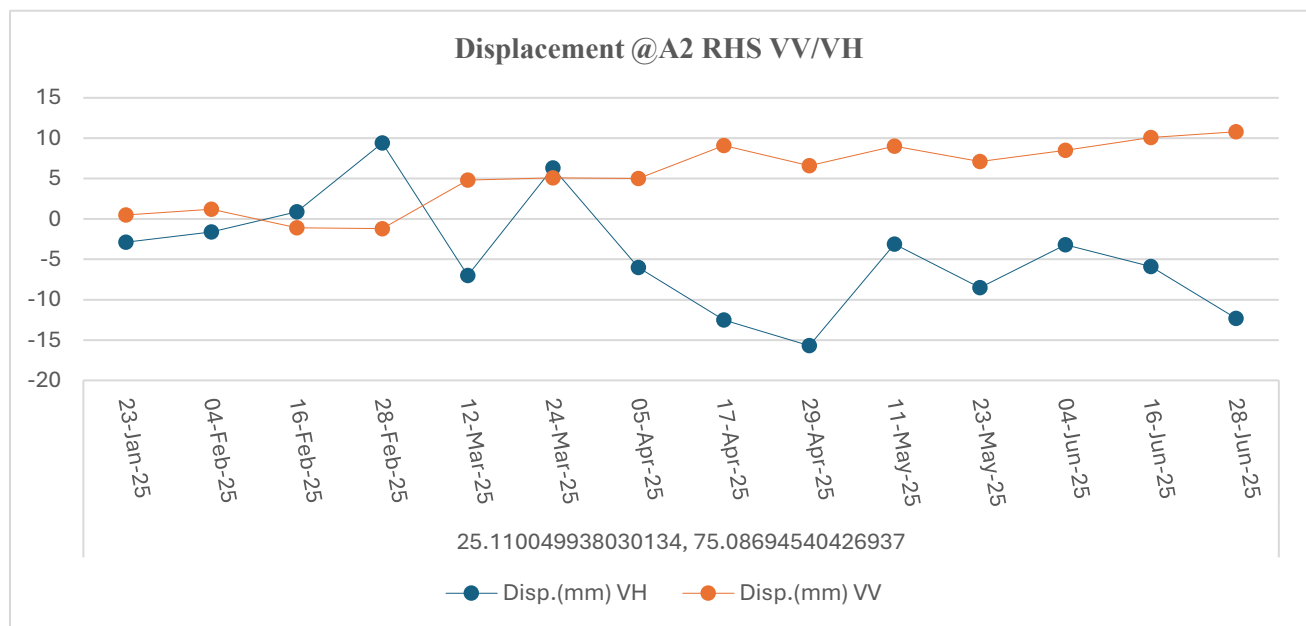
Critical Events & Anomalies:

- **February 2025:** VH shows an initial notable rise to about +6 mm, marking the beginning of sustained positive lateral movement while VV remains moderately negative.
- **June 2025:** VH records its highest positive displacement (+15 mm), while VV again drops toward -17 mm, highlighting intensified horizontal movement with renewed vertical decline

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.110049938030134, 75.08694540426937)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical displacement (VV)** Vertical displacement shows a clear and consistent upward trend throughout the monitoring period. VV values begin near 0 to +1 mm in January and gradually increase to approximately +10 to +12 mm by late June.
- **Horizontal displacement (VH)** Horizontal displacement exhibits high variability with multiple directional changes, though it is predominantly negative after March 2025.

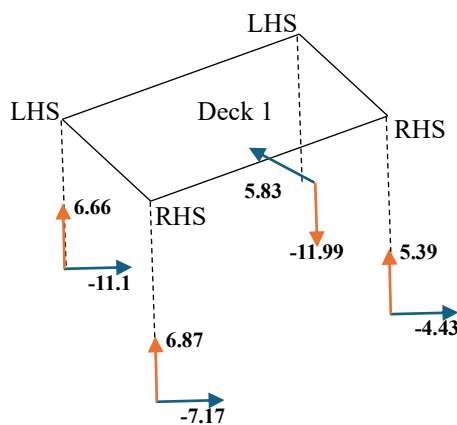
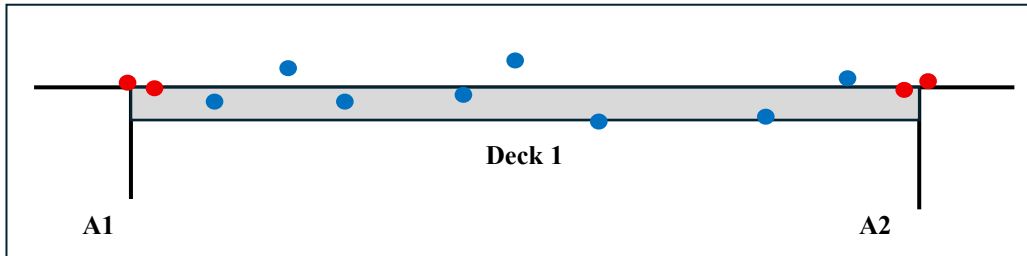
Critical Events & Anomalies:

- **February 2025:** VH reaches a positive peak of about +9 mm, representing the maximum positive lateral displacement before the trend reverses.
- **June 2025:** VV attains its highest uplift values (+10 to +12 mm) while VH remains negative (−5 to −12 mm), indicating simultaneous vertical uplift with continued lateral displacement.

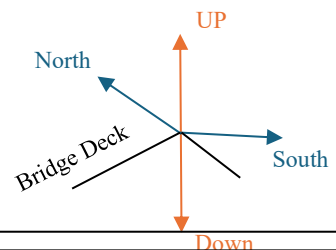
ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis



Vertical Displacement Behaviour:

- The deck slab exhibits moderate differential vertical displacement, with values ranging approximately from -11.99 mm (downward) to $+6.87$ mm (upward) across LHS and RHS locations. The RHS of the deck shows notable downward movement (-11.99 mm) at one monitoring point, indicating localized settlement or compression effects

Horizontal Displacement Behaviour:

- Horizontal displacements are pronounced and directionally varied, with magnitudes ranging from approximately -11.1 mm to $+5.83$ mm. The LHS and central RHS points show southward movement (up to -11.1 mm), while the deck top indicates a northward component ($+5.83$ mm). The RHS outer point records a smaller but distinct southward displacement (-4.43 mm), reinforcing asymmetric lateral behaviour across the deck width

CONCLUSION:

- The combined vertical and horizontal displacement patterns indicate that Deck 1 is undergoing torsional deformation, characterized by simultaneous rotation and warping of the slab. The asymmetry between LHS and RHS responses, both vertically and horizontally, suggests eccentric load distribution, differential bearing response, or thermal effects as contributing factors.